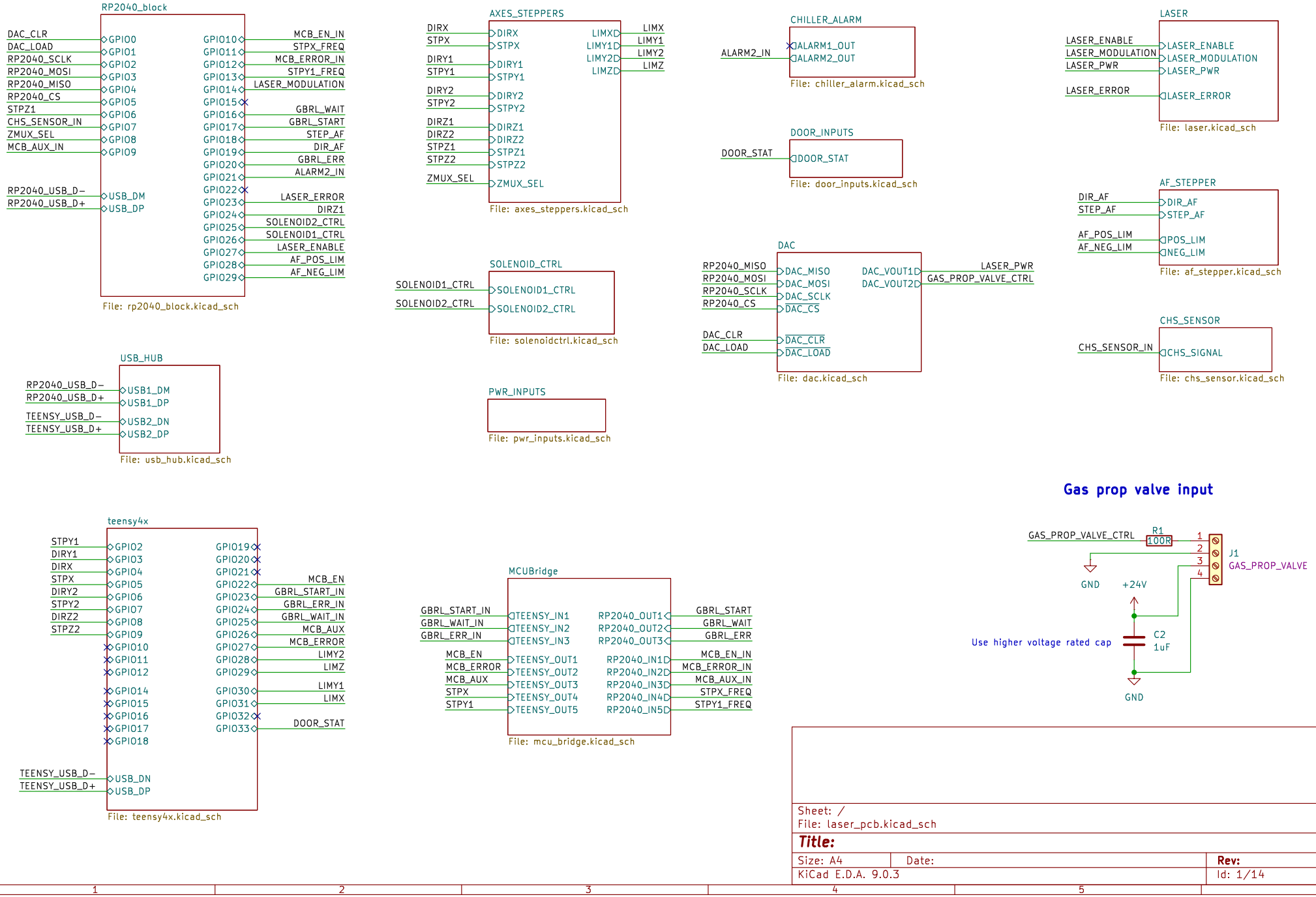
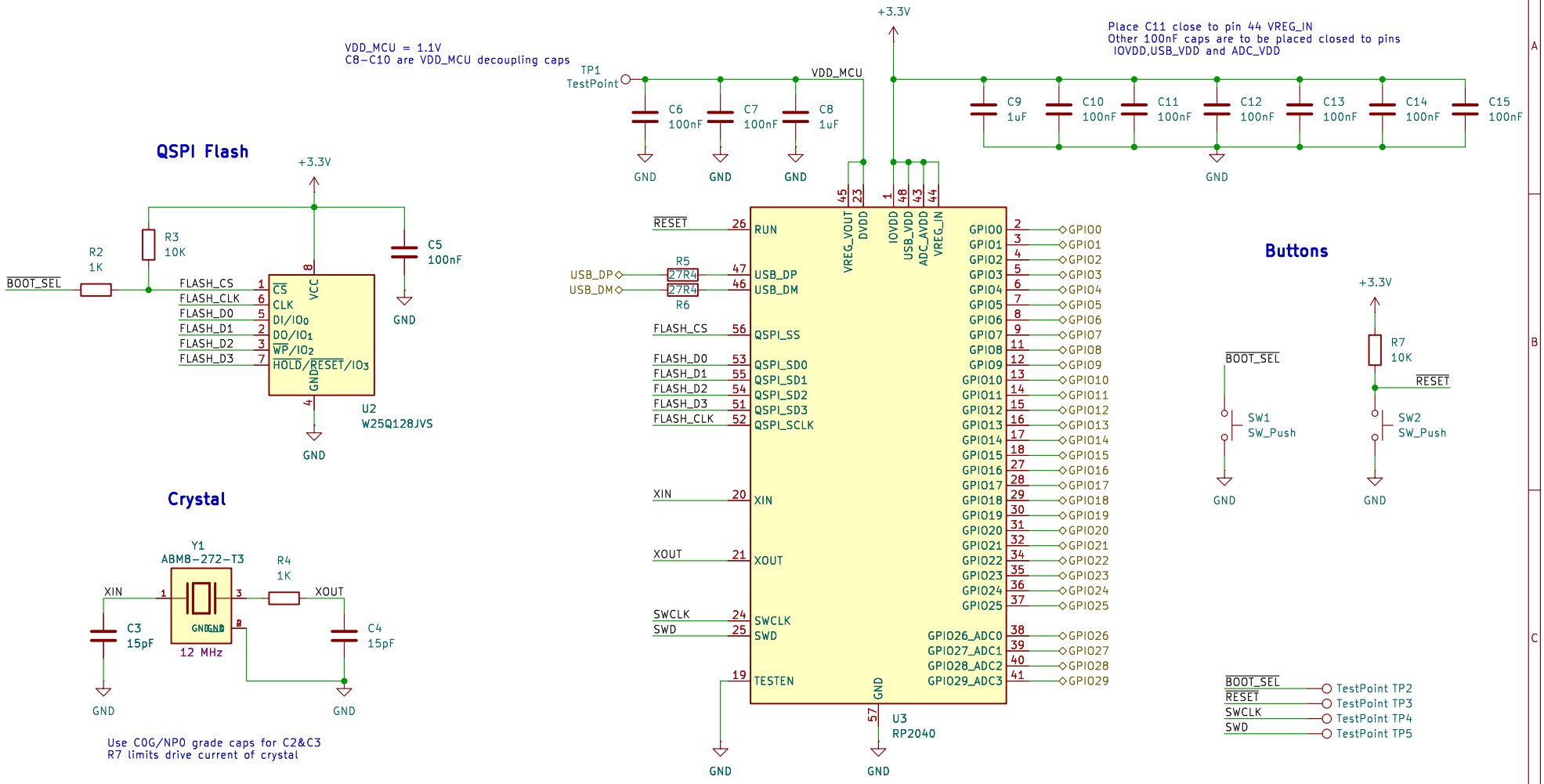

RP2040_block
DAC_CLR
DAC_LOAD
RP2040_SCLK
RP2040_MOSI
RP2040_MISO
RP2040_CS
STPZ1
CHS_SENSOR_IN
ZMUX_SEL
MCB_AUX_IN
RP2040_USB_D−
RP2040_USB_D+
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
USB_DM
USB_DP
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
GPIO22
GPIO23
GPIO24
GPIO25
GPIO26
GPIO27
GPIO28
GPIO29
MCB_EN_IN
STPX_FREQ
MCB_ERROR_IN
STPY1_FREQ
LASER_MODULATION
GBRL_WAIT
GBRL_START
STEP_AF
DIR_AF
GBRL_ERR
ALARM2_IN
LASER_ERROR
DIRZ1
SOLENOID2_CTRL
SOLENOID1_CTRL
LASER_ENABLE
AF_POS_LIM
AF_NEG_LIM
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AXES_STEPPERS
DIRX
STPX
DIRY1
STPY1
DIRY2
STPY2
DIRZ1
DIRZ2
STPZ1
STPZ2
ZMUX_SEL
LIMX
LIMY1
LIMY2
LIMZ
File: axes_steppers.kicad_sch
CHILLER_ALARM
ALARM2_IN
ALARM1_OUT
ALARM2_OUT
File: chiller_alarm.kicad_sch
DOOR_INPUTS
DOOR_STAT
File: door_inputs.kicad_sch
LASER
LASER_ENABLE
LASER_MODULATION
LASER_PWR
LASER_ERROR
File: laser.kicad_sch
AF_STEPPER
DIR_AF
STEP_AF
AF_POS_LIM
AF_NEG_LIM
POS_LIM
NEG_LIM
File: af_stepper.kicad_sch
SOLENOID_CTRL
SOLENOID1_CTRL
SOLENOID2_CTRL
File: solenoidctrl.kicad_sch
DAC
RP2040_MISO
RP2040_MOSI
RP2040_SCLK
RP2040_CS
DAC_CLR
DAC_LOAD
DAC_MISO
DAC_MOSI
DAC_SCLK
DAC_CS
DAC_VOUT1
DAC_VOUT2
LASER_PWR
GAS_PROP_VALVE_CTRL
File: dac.kicad_sch
CHS_SENSOR
CHS_SENSOR_IN
CHS_SIGNAL
File: chs_sensor.kicad_sch
USB_HUB
RP2040_USB_D−
RP2040_USB_D+
TEENSY_USB_D−
TEENSY_USB_D+
USB1_DM
USB1_DP
USB2_DN
USB2_DP
File: usb_hub.kicad_sch
PWR_INPUTS
File: pwr_inputs.kicad_sch
Gas prop valve input
GAS_PROP_VALVE_CTRL
R1
100R
GND
+24V
Use higher voltage rated cap
C2
1uF
J1
GAS_PROP_VALVE
teensy4x
STPY1
DIRY1
DIRX
STPX
DIRY2
STPY2
DIRZ2
STPZ2
TEENSY_USB_D−
TEENSY_USB_D+
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO14
GPIO15
GPIO16
GPIO17
GPIO18
USB_DN
USB_DP
GPIO19
GPIO20
GPIO21
GPIO22
GPIO23
GPIO24
GPIO25
GPIO26
GPIO27
GPIO28
GPIO29
GPIO30
GPIO31
GPIO32
GPIO33
MCB_EN
GBRL_START_IN
GBRL_ERR_IN
GBRL_WAIT_IN
MCB_AUX
MCB_ERROR
LIMY2
LIMZ
LIMY1
LIMX
DOOR_STAT
File: teensy4x.kicad_sch
MCUBridge
GBRL_START_IN
GBRL_WAIT_IN
GBRL_ERR_IN
MCB_EN
MCB_ERROR
MCB_AUX
STPX
STPY1
TEENSY_IN1
TEENSY_IN2
TEENSY_IN3
TEENSY_OUT1
TEENSY_OUT2
TEENSY_OUT3
TEENSY_OUT4
TEENSY_OUT5
RP2040_OUT1
RP2040_OUT2
RP2040_OUT3
RP2040_IN1
RP2040_IN2
RP2040_IN3
RP2040_IN4
RP2040_IN5
GBRL_START
GBRL_WAIT
GBRL_ERR
MCB_EN_IN
MCB_ERROR_IN
MCB_AUX_IN
STPX_FREQ
STPY1_FREQ
File: mcu_bridge.kicad_sch
Sheet: /
File: laser_pcb.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.3
Id: 1/14

# RP2040 MCU




| | |
|---|---|
| Sheet: /RP2040_block/ | |
| File: rp2040_block.kicad_sch | |
| **Title:** | |
| Size: A4 | Date: |
| KiCad E.D.A. 9.0.3 | Rev: Id: 2/14 |

# GBRL Board Main MCU — MIMXRT1062

HF crystal, 24MHz, CL = 8pF, ESR = 50ohm
Use NP0/C0G grade caps.

Adjust cap values if
stray capacitance ends up being
more/less than expected.

Series resistor in front of XTALO
can be replaced with non zero value.
This limit crystal drive current
if needed

Y3
24.000MHz
CS14522-24M
XTALI
XTALO
GND
R8 2.2M
C17 10pF
C19 10pF

R9 0R
ABS07-32.768KHZ-T
Y2 32.768kHz
C16 18pF
C18 18pF

LF crystal, 32.768kHz
CL = 12.5pF, ESR = 70kOhm
Use NP0/C0G grade caps.

Adjust cap values if
stray capacitance ends up being
more/less than expected.

TP6 TestPoint ONOFF
TP9 TestPoint RX1
TP10 TestPoint TX1

U4B
MIMXRT1062DVJ6B

| | | |
|---|---|---|
| | VDDA_ADC_3P3 | 3V3 |
| | VDD_HIGH_CAP | VDD_HIGH_CAP |
| | VDD_HIGH_IN | 3V3 |
| | VDD_SNVS_CAP | VDD_SNVS_CAP |
| | VDD_SNVS_IN | VDD_SNVS_IN |
| | VDD_SOC_IN | VDD_SOC |
| | VDD_SOC_IN_1 | |
| | VDD_SOC_IN_2 | |
| | VDD_SOC_IN_3 | |
| | VDD_SOC_IN_4 | |
| | VDD_SOC_IN_5 | |
| | VDD_SOC_IN_6 | |
| | VDD_SOC_IN_7 | |
| | VDD_SOC_IN_8 | |
| | VDD_USB_CAP | VDD_USB_CAP |
| | DCDC_IN | DCDC_IN |
| | DCDC_IN_1 | |
| | DCDC_IN_Q | 3V3 |
| | DCDC_LP | DCDC_LP |
| | DCDC_LP_1 | |
| | DCDC_PSWITCH | PTA3 |
| | DCDC_SENSE | VDD_SOC |
| | NVCC_EMC | 3V3 |
| | NVCC_EMC_1 | |
| | NVCC_GPIO | |
| | NVCC_GPIO_1 | |
| | NVCC_GPIO_2 | |
| | NVCC_PLL | NVCC_PLL |
| | NVCC_SD0 | 3V3 |
| | NVCC_SD1 | |
| | GPANAIO | |
| | PMIC_ON_REQ | PMIC_ON_REQ |
| | PMIC_STBY_REQ | |
| | USB_OTG1_VBUS | USB_VBUS_1 |
| | USB_OTG1_CHD_B | |
| | USB_OTG1_DP | USBD_P |
| | USB_OTG1_DN | USBD_N |
| | USB_OTG2_VBUS | |
| | USB_OTG2_DP | |
| | USB_OTG2_DN | |
| CCM_CLK1_P | | |
| CCM_CLK1_N | | |
| XTALI | | |
| XTALO | | |
| RTC_XTALI | | |
| RTC_XTALO | | |
| ONOFF | | |
| PTB1 | POR_B | |
| TEST_MODE | | |
| WAKEUP | | |
| GPIO_EMC_00 | | |
| PTA1 | GPIO_EMC_01 | |
| GPIO_EMC_02 | | |
| GPIO_EMC_03 | | |
| EMC_04 | GPIO_EMC_04 | |
| EMC_05 | GPIO_EMC_05 | |
| EMC_06 | GPIO_EMC_06 | |
| EMC_07 | GPIO_EMC_07 | |
| EMC_08 | GPIO_EMC_08 | |
| GPIO_EMC_09 | | |
| GPIO_EMC_10 | | |
| GPIO_EMC_11 | | |
| GPIO_EMC_12 | | |
| GPIO_EMC_13 | | |
| GPIO_EMC_14 | | |
| GPIO_EMC_15 | | |
| GPIO_EMC_16 | | |
| GPIO_EMC_17 | | |
| GPIO_EMC_18 | | |
| GPIO_EMC_19 | | |
| GPIO_EMC_20 | | |
| GPIO_EMC_21 | | |
| GPIO_EMC_22 | | |
| GPIO_EMC_23 | | |
| GPIO_EMC_24 | | |
| GPIO_EMC_25 | | |
| GPIO_EMC_26 | | |
| GPIO_EMC_27 | | |
| GPIO_EMC_28 | | |
| GPIO_EMC_29 | | |
| GPIO_EMC_30 | | |
| GPIO_EMC_31 | | EMC_31 |
| GPIO_EMC_32 | | EMC_32 |
| GPIO_EMC_33 | | |
| GPIO_EMC_34 | | |
| GPIO_EMC_35 | | |
| GPIO_EMC_36 | | EMC_36 |
| GPIO_EMC_37 | | EMC_37 |
| GPIO_EMC_38 | | |
| GPIO_EMC_39 | | |
| GPIO_EMC_40 | | |
| GPIO_EMC_41 | | |
| NGND_KEL0 | | |
| DCDC_GND | | |
| DCDC_GND_1 | | |
| VSS | | |
| VSS_1 | | |
| VSS_2 | | |
| VSS_3 | | |
| VSS_4 | | |
| VSS_5 | | |
| VSS_6 | | |
| VSS_7 | | |
| VSS_8 | | |
| VSS_9 | | |
| VSS_10 | | |
| VSS_11 | | |
| VSS_12 | | |
| VSS_13 | | |
| VSS_14 | | |
| VSS_15 | | |
| VSS_16 | | |
| VSS_17 | | |
| VSS_18 | | |

U4A
MIMXRT1062DVJ6B

| | | |
|---|---|---|
| TX1 | GPIO_AD_B0_00 | GPIO_AD_B1_00 AD_B1_00 |
| RX1 | GPIO_AD_B0_01 | GPIO_AD_B1_01 AD_B1_01 |
| PTA4 | GPIO_AD_B0_02 | GPIO_AD_B1_02 AD_B1_02 |
| | GPIO_AD_B0_03 | GPIO_AD_B1_03 AD_B1_03 |
| | GPIO_AD_B0_04 | GPIO_AD_B1_04 |
| PTA5 | GPIO_AD_B0_05 | GPIO_AD_B1_05 |
| PTB0 | GPIO_AD_B0_06 | GPIO_AD_B1_06 AD_B1_06 |
| PTB5 | GPIO_AD_B0_07 | GPIO_AD_B1_07 AD_B1_07 |
| PTA7 | GPIO_AD_B0_08 | GPIO_AD_B1_08 AD_B1_08 |
| PTA6 | GPIO_AD_B0_09 | GPIO_AD_B1_09 AD_B1_09 |
| PTA3 | GPIO_AD_B0_10 | GPIO_AD_B1_10 AD_B1_10 |
| AD_B0_12 | GPIO_AD_B0_11 | GPIO_AD_B1_11 AD_B1_11 |
| AD_B0_13 | GPIO_AD_B0_12 | GPIO_AD_B1_12 |
| | GPIO_AD_B0_13 | GPIO_AD_B1_13 |
| | GPIO_AD_B0_14 | GPIO_AD_B1_14 AD_B1_14 |
| | GPIO_AD_B0_15 | GPIO_AD_B1_15 AD_B1_15 |
| B0_00 | GPIO_B0_00 | GPIO_B1_00 B1_00 |
| B0_01 | GPIO_B0_01 | GPIO_B1_01 B1_01 |
| B0_02 | GPIO_B0_02 | GPIO_B1_02 |
| | GPIO_B0_03 | GPIO_B1_03 |
| | GPIO_B0_04 | GPIO_B1_04 |
| | GPIO_B0_05 | GPIO_B1_05 |
| | GPIO_B0_06 | GPIO_B1_06 |
| | GPIO_B0_07 | GPIO_B1_07 |
| | GPIO_B0_08 | GPIO_B1_08 |
| | GPIO_B0_09 | GPIO_B1_09 |
| B0_10 | GPIO_B0_10 | GPIO_B1_10 |
| B0_11 | GPIO_B0_11 | GPIO_B1_11 |
| B0_12 | GPIO_B0_12 | GPIO_B1_12 |
| PTB4 | GPIO_B0_13 | GPIO_B1_13 |
| | GPIO_B0_14 | GPIO_B1_14 |
| | GPIO_B0_15 | GPIO_B1_15 |
| | GPIO_SD_B0_00 | GPIO_SD_B1_00 |
| | GPIO_SD_B0_01 | GPIO_SD_B1_01 |
| | GPIO_SD_B0_02 | GPIO_SD_B1_02 |
| | GPIO_SD_B0_03 | GPIO_SD_B1_03 |
| | GPIO_SD_B0_04 | GPIO_SD_B1_04 |
| | GPIO_SD_B0_05 | GPIO_SD_B1_05 |
| | | GPIO_SD_B1_06 FLASH_CS |
| | | GPIO_SD_B1_07 FLASH_CLK |
| | | GPIO_SD_B1_08 FLASH_D0 |
| | | GPIO_SD_B1_09 FLASH_D1 |
| | | GPIO_SD_B1_10 FLASH_D2 |
| | | GPIO_SD_B1_11 FLASH_D3 |

B0_00 B0_01 B0_02 470R R13 D1 LED_ORANGE GND

USB_DP 27R4 R10 USBD_P
USB_DN 27R4 R11 USBD_N

J28 TEENSY_USB_DEBUG
R42 R43 USB_DP USB_DN

GPIO02 EMC_04
GPIO03 EMC_05
GPIO04 EMC_06
GPIO05 EMC_08
GPIO06 B0_10
GPIO07 B1_01
GPIO08 B1_00
GPIO09 B0_11
GPIO10 B0_00
GPIO11 B0_02
GPIO12 B0_01
GPIO14 AD_B1_02
GPIO15 AD_B1_03
GPIO16 AD_B1_07
GPIO17 AD_B1_06
GPIO18 AD_B1_01
GPIO19 AD_B1_00
GPIO20 AD_B1_10
GPIO21 AD_B1_11
GPIO22 AD_B1_08
GPIO23 AD_B1_09
GPIO24 AD_B0_12
GPIO25 AD_B0_13
GPIO26 AD_B1_14
GPIO27 AD_B1_15
GPIO28 EMC_32
GPIO29 EMC_31
GPIO30 EMC_37
GPIO31 EMC_36
GPIO32 B0_12
GPIO33 EMC_07

GPIO numbering here is referenced to Teensy 4.1

## Bootloader MCU — MKL02Z32VFG4

C20 4.7uF
C21 0.22uF
3V3
GND

U5
MKL02Z32VFG4

| | | |
|---|---|---|
| | VDD/VREFH | |
| PTA1 | PTA0/IRQ_0 | |
| | PTA1/IRQ_1/LPTMR0_ALT1 | |
| PTA3 | PTA2 | |
| PTA4 | PTA3 | |
| PTA5 | PTA4 | |
| PTA6 | PTA5 | |
| PTA7 | PTA6 | |
| | PTA7/IRQ_4 | |
| | PTB0/IRQ_5 | PTB0 |
| | PTB1/IRQ_6 | PTB1 |
| | PTB2/IRQ_7 | |
| | PTB3/IRQ_10 | |
| | PTB4/IRQ_11 | PTB4 |
| | PTB5/IRQ_12 | PTB5 |
| | EPAD | |
| | VSS/VREFL | |

R12 100K GND

PROG button
SW3 SW_Push
R14 470R
3V3
D2 LED_RED

Teensy bootloader allows
usage of button to enter programming
mode.

PROG_LED lights up to indicate
programming status and faults

## MIMXRT1062 power rails passives

VDD_SNVS_IN needs to first be powered up.
VDD_USB_CAP gets power from USB_OTG1_VBUS
and provides a 3.0V source to VDD_SNVS_IN.

Diode array is to let the 3.3V rail power
VDD_SNVS_IN once the regulator is enabled
(once PMIC_ON_REQ is driven HIGH by MCU)

Once 3.3V rail stabilizes, DCDC_PSWITCH is driven
by bootloader MCU to enable the internal DC-DC converter
on main MCU.

This provides the 1.1V VDD SOC rail output.

All power rail caps must be placed as close as
possible to power pads on MCU.

VDD_SNVS_CAP C31 0.22uF GND
VDD_HIGH_CAP C34 0.22uF C38 4.7uF GND GND

L1 4.7uH
DCDC_LP VDD_SOC TP16 TestPoint
C28 10uF C32 4.7uF C35 0.22uF C39 10nF GND

3V3
TP11 TestPoint VDD_SNVS_IN D3 BAT54C
C22 0.22uF GND

3V3
FB3 BLM18KG121TN1D
VDD_USB_CAP DCDC_IN NVCC_PLL
C23 10uF C24 4.7uF C26 0.22uF
C29 10uF C33 4.7uF C36 0.22uF
C40 0.22uF C41 4.7uF
GND GND GND

## 3.3V regulator

+5V
R15 0R
FB2 BLM18KG121TN1D
F1 0805L050WR 500mA
TP13 TestPoint USB_VBUS_1
C25 2.2uF C27 0.22uF C30 0.22uF GND

U7
TLV75733PDRV
IN OUT EN GND
3V3 TP14 TestPoint
R18 100K
PMIC_ON_REQ TP15 TestPoint

Max input voltage: 5.5V
Max output load current: 1A
F1 limits current to 500mA

PMIC_ON_REQ signal from MCU turns
on 3.3V regulator. This is important
since MCU has specific power up sequence,
so 3.3V regulator cannot be turned on immediately.

C30 is to be placed close to the MCU.

## QSPI Flash

3V3
R17 10K
C37 0.22uF
GND

FLASH_CS
FLASH_CLK
FLASH_D0
FLASH_D1
FLASH_D2
FLASH_D3

U6
W25Q64JVSSIQ
CS CLK DI/IO0 DO/IO1 WP/IO2 HOLD/RESET/IO3 VCC GND
GND

R17 is extra-provisioned
CS pullup.

It can be removed after testing
if not actually required. Otherwise
if having issues with flash, leave it in,

bootloader chip was programmed to use flash chips
ending in code "IM"

Sheet: /teensy4x/
File: teensy4x.kicad_sch
**Title:**
Size: A3 | Date: | **Rev:**
KiCad E.D.A. 9.0.3 | Id: 4/14

## USB Hub Controller

NON_REM[1:0] = 10
This is to denote that both connected
downstream devices are permanently attached.

This allows hub controller to notify connected
host that downstream devices are not hot-pluggable.

+3.3V
R76 50K
R78 50K
NON_REM0
NON_REM1
R77 50K
R79 50K
GND
GND
CFG_SEL0
R80 50K
GND
VBUS_IN
R74 100K
R75 100K
GND

U28
USB2422T-I/MJ

| | | | |
|---|---|---|---|
| 16 | VBUS_DET | PRTPWR1*(BC_EM1) | 7 |
| USB_IN_D+ 20 | USBDP_UP | OCS1_N | 8 |
| USB_IN_D- 19 | USBDM_UP | USBDP_DN1*PART_DIS_P1 | 3 → USB1_DP |
| | | USBDM_DN1*PART_DIS_M1 | 2 → USB1_DM |
| CFG_SEL0 14 | SMBCLK*CFG_SEL0 | | |
| NON_REM1 13 | SMBDATA*NON_REM1 | PRTPWR2 | 11 |
| 15 | RESET_N | OCS2_N | 12 |
| NON_REM0 17 | SUSP_IND*LOCAL_PWR*(NON_REM0) | USBDP_DN1*PART_DIS_P2 | 5 → USB2_DP |
| | | USBDM_DN1*PART_DIS_M2 | 4 → USB2_DN |
| XTALI 22 | XTALIN*CLKIN | | |
| XTALO 21 | XTAOUT*(CLKIN_EM) | VDD33 | 1 |
| | | VDD33 | 9 |
| | | VDD33 | 18 |
| | | CRFILT | 10 |
| | | PLLFILT | 23 |
| | | RBIAS | 24 |
| | NC 6 | EPAD/GND 25 | |

TP44 TestPoint
TP45 TestPoint
TP46 TestPoint

Y4
24.000MHz
XTALI
XTALO
R73
C87 10pF
C88 10pF
GND

HF crystal, 24MHz, CL = 8pF, ESR = 50ohm
Use NPO/C0G grade caps.

Adjust C40 and C41 values if
stray capacitance ends up being
more/less than expected.

R81 12K
C89 100nF
C90 100nF

Use 1% tolerance part for R78

Hub controller has termination resistors
built in.

+3.3V
C91 1uF
C92 100nF
GND

C92 should be placed close to pin 9
C93 should be placed close to pin 1

## USB 2.0 Type-C connector

J26
USB4105-GF-A

| | | |
|---|---|---|
| VBUS | A4 | |
| CC1 | A5 | CC1 |
| CC2 | B5 | CC2 |
| D- | A7 | |
| D- | B7 | |
| D+ | A6 | |
| D+ | B6 | |
| SBU1 | A8 | |
| SBU2 | B8 | |
| SHIELD | | |
| GND | | |
| S1 | | |
| A1 | | |

FB5
BLM18KG121TN1D
GND

BLM18KG121TN1D
FB6
VBUS_IN
TP47 TestPoint
C93 100nF
GND

U29
USBLC6-2SC6
1 → 6 USB_IN_D-
3 → 4 USB_IN_D+
5
2
GND

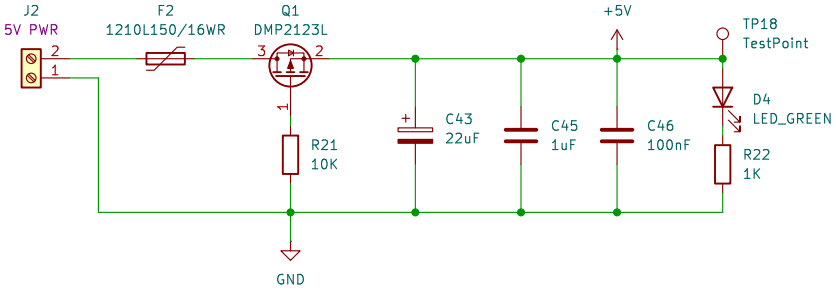
CC1
R82 5.1K
GND
CC2
R83 5.1K
GND

Sheet: /USB_HUB/
File: usb_hub.kicad_sch

**Title:**

| | | |
|---|---|---|
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 9.0.3 | | Id: 13/14 |

## +5V rail input

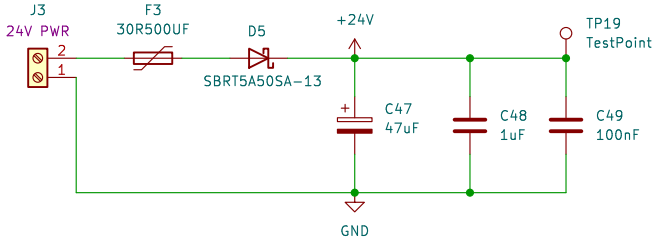



F2 provides short-circuit & overcurrent protection.
Hold current =1.5A, Trip current=3A

Fuse can be bypassed or replaced with suitable alternative.

Q1 acts as a near "ideal-diode" for reverse polarity protection.

## +24V rail input

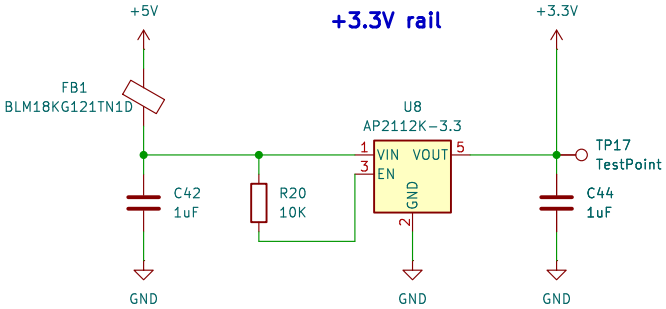



F3 provides short-circuit & overcurrent protection
Hold current = 5A, Trip current = 10A

D4 - low drop reverse voltage protection

!Use higher voltage rated caps for 24V rail

## +3.3V rail




Max input voltage: 6.5V
Max output load current: 600mA

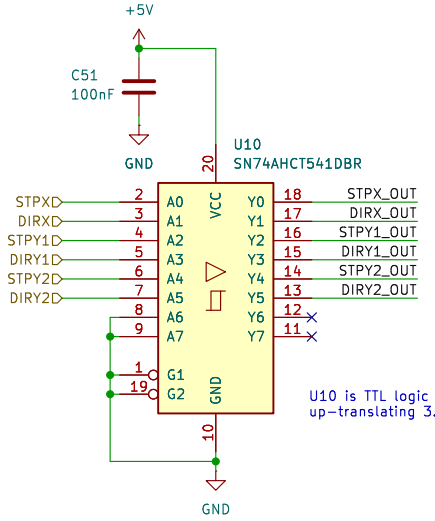
EN pulled HIGH to permanently enable regulator

This rail provides 3.3V to the whole board except for the Teesny section, which has its own 3V3 regulation

FB1 can be removed if required.

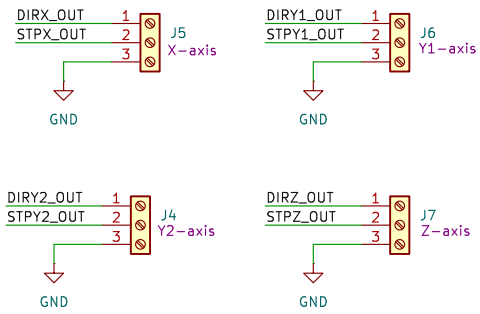
| | | |
|---|---|---|
| Sheet: /PWR_INPUTS/ | | |
| File: pwr_inputs.kicad_sch | | |
| **Title:** | | |
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 9.0.3 | | Id: 4/14 |

## X,Y1,Y2 axes steppers control signals driver

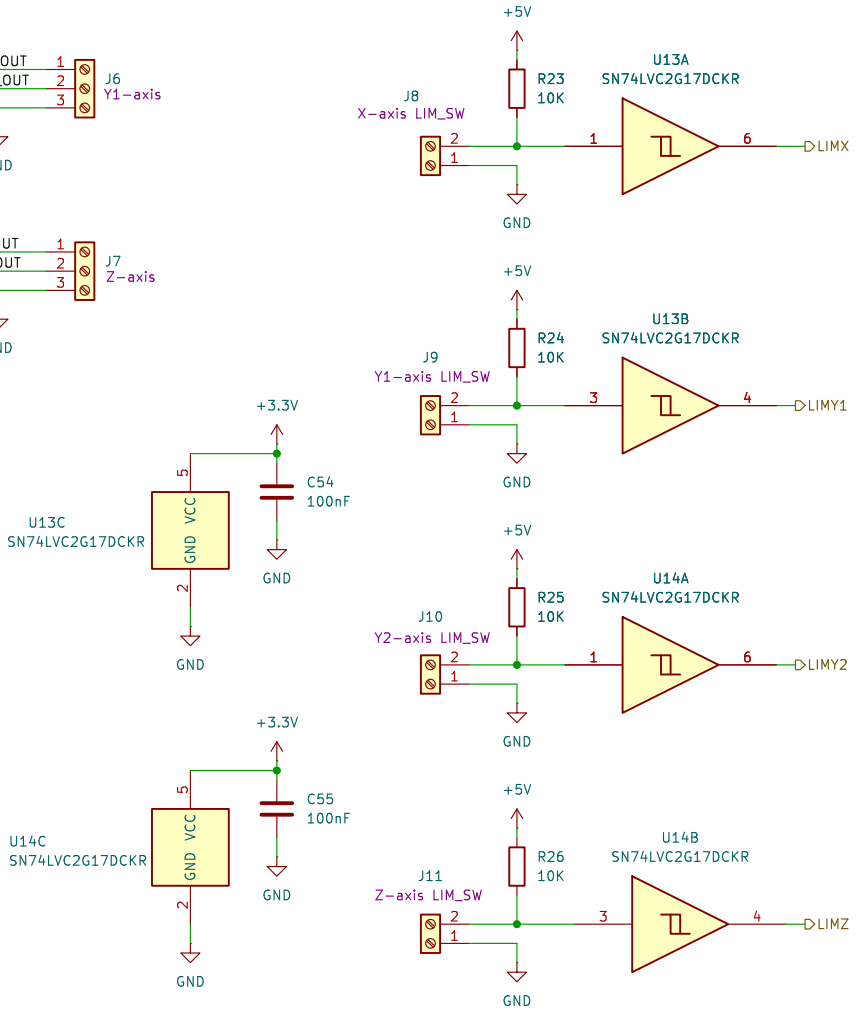



U10 is TTL logic compatible, effectively up-translating 3.3V -> 5V

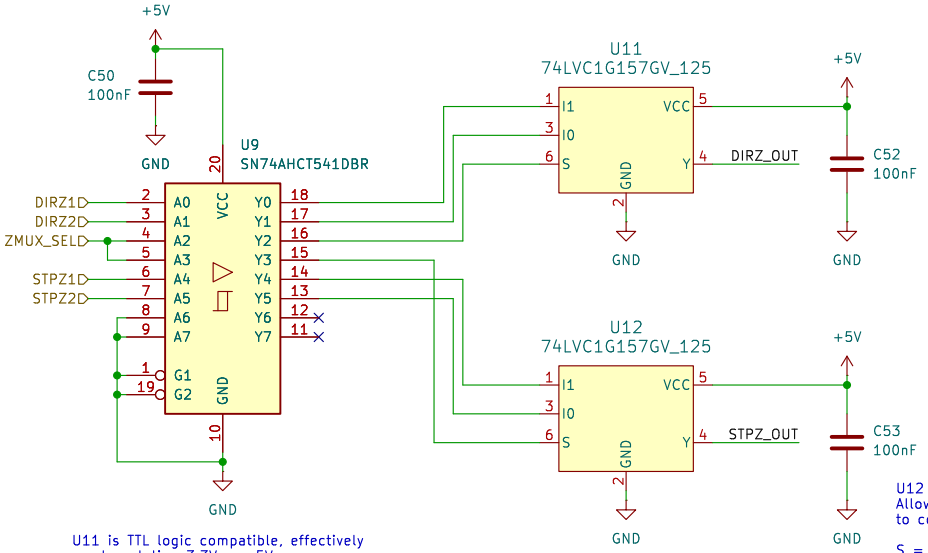
## Stepper motor connectors




## Stepper axes limit switch connectors




## Z axis stepper control signal driver & muxing




U11 is TTL logic compatible, effectively up-translating 3.3V -> 5V

U12 and U13 are signal muxes. Allows both RP2040 and Teensy to control Z axis DIR & STEP lines

S = LOW, I0 is selected (DIRZ2,STPZ2)
S = HIGH, I1 is selected (DIRZ1,STPZ1)
S on both muxes are controlled by input ZMUX_SEL

Sheet: /AXES_STEPPERS/
File: axes_steppers.kicad_sch

**Title:**

| Size: A4 | Date: | Rev: |
|---|---|---|
| KiCad E.D.A. 9.0.3 | | Id: 5/14 |

## High-side P-FET switches for solenoids

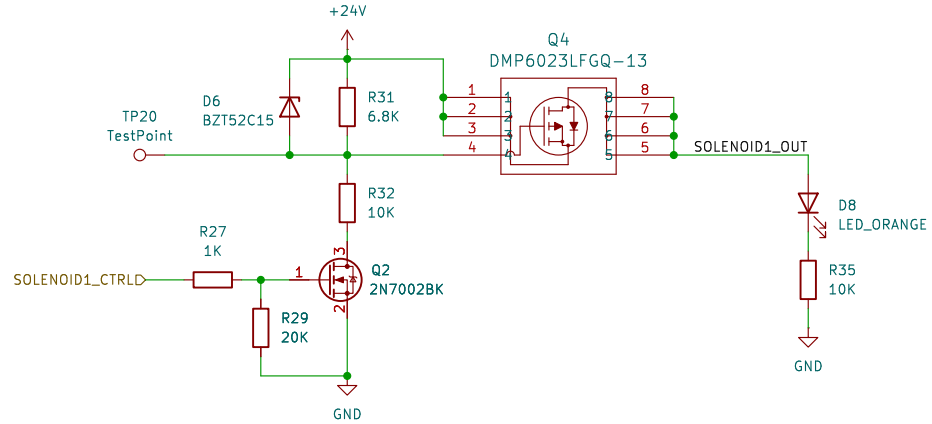
Q3 is a PMOS, allows high side switching
Max current rating: 6.8-7A
Max Vgs = -20V

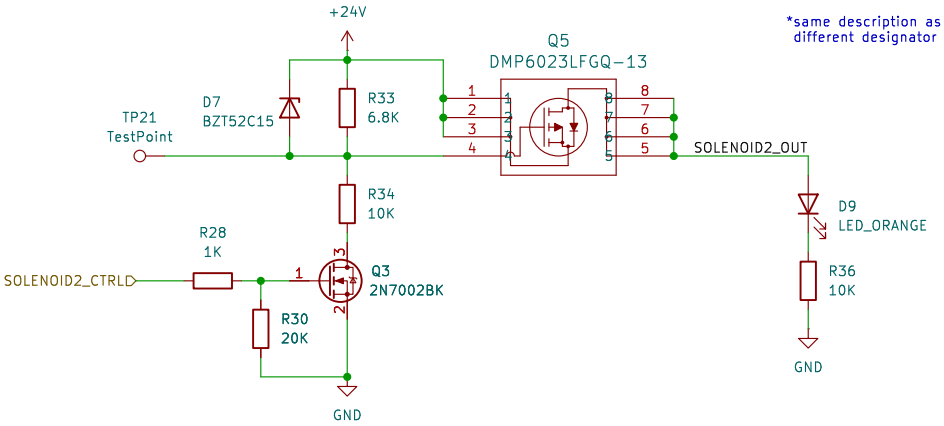
R20 and R21 form a voltage divider to limit
Vgs to around -9.7V. This ensures Q3 is very close
to the max optimized Vgs level of -10V.

D6 is a 15V Zener, clamps Vgs to -14 to -15.5V max,
in case of transients.

Q4 is an N-FET, drives Q3's gate LOW when active
and vice versa, allowing for active high logic.

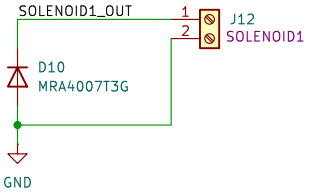
D7 is a visual indicator of when P-FET is active,
allowing easy troubleshooting

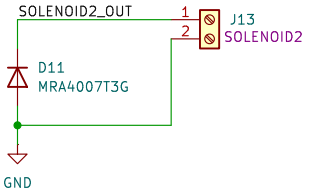



*same description as above - just
different designator numbers

## Solenoid input connectors

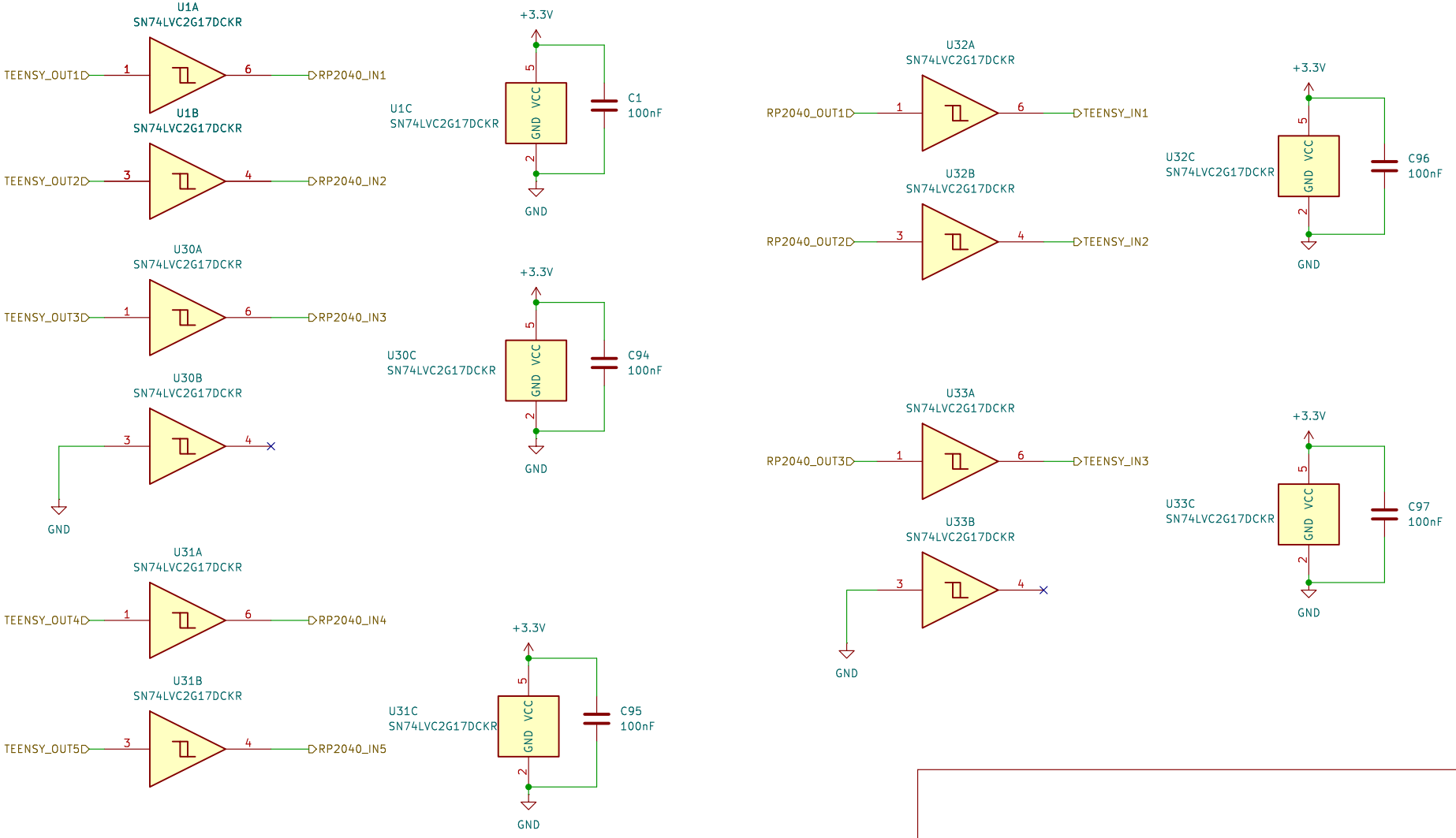



D10 and D11 act as flyback diodes,
protecting P-FET drain from inductive kickback
from solenoid coils, once turned off

| | | |
|---|---|---|
| Sheet: /SOLENOID_CTRL/ | | |
| File: solenoidctrl.kicad_sch | | |
| **Title:** | | |
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 9.0.3 | | Id: 6/14 |

# RP2040 <–> Teensy signals buffer




| | | |
|---|---|---|
| Sheet: /MCUBridge/ | | |
| File: mcu_bridge.kicad_sch | | |
| **Title:** | | |
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 9.0.3 | | Id: 14/14 |

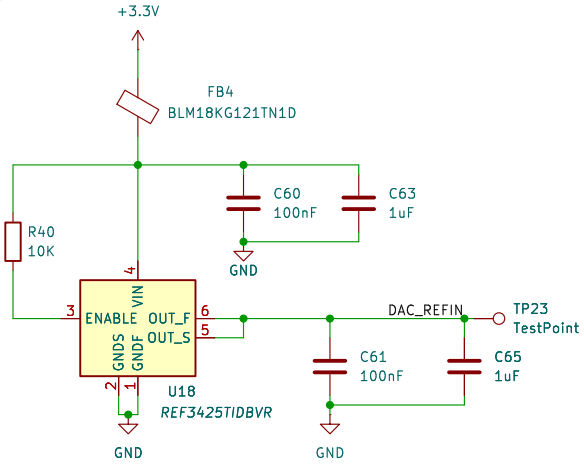
## DAC +2.5V reference

VIN can be powered by +3.3 or +5V, default is 3.3V.

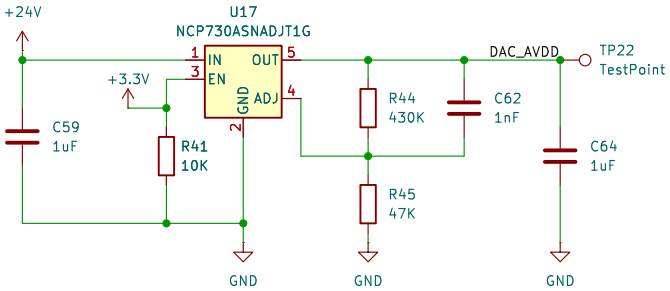
EN tied to VIN to permanently enable reference.

REF output = +2.5V

Kelvin outputs of reference not required as DAC output current would be very low.




## DAC analog power input rail




U18 provides power to DAC's AVDD input.

If +3.3V rail is missing, then EN gets pulled LOW. This means that + 3.3V rail must be enabled to turn on power to the DAC even though Vin is present.

DAC_AVDD = ~ 12.2V

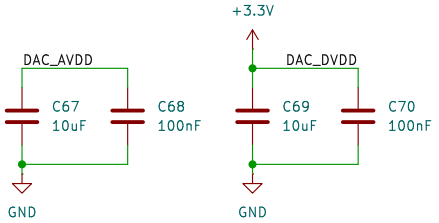
Vout = 1.2(R44/R45 + 1) + (Iadj * R44)
Iadj worst case can be taken as 0.1uA.

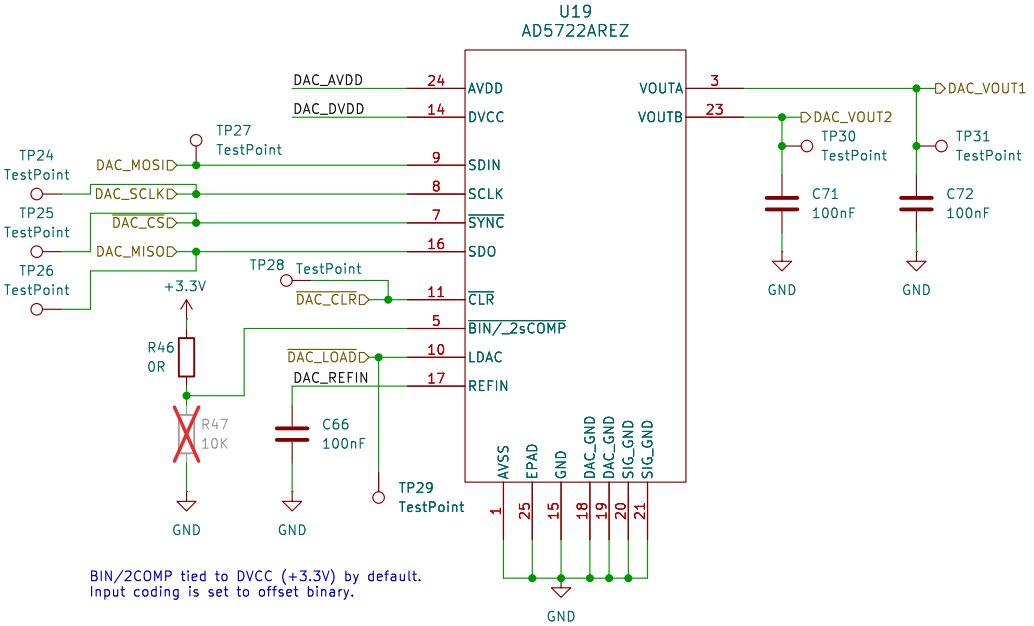
C64 provides filtering to feedback input.

Use 1% or better tolerances for R45 and R46

For more info, refer to U18's datasheet

## AD5722 – 2 channel DAC

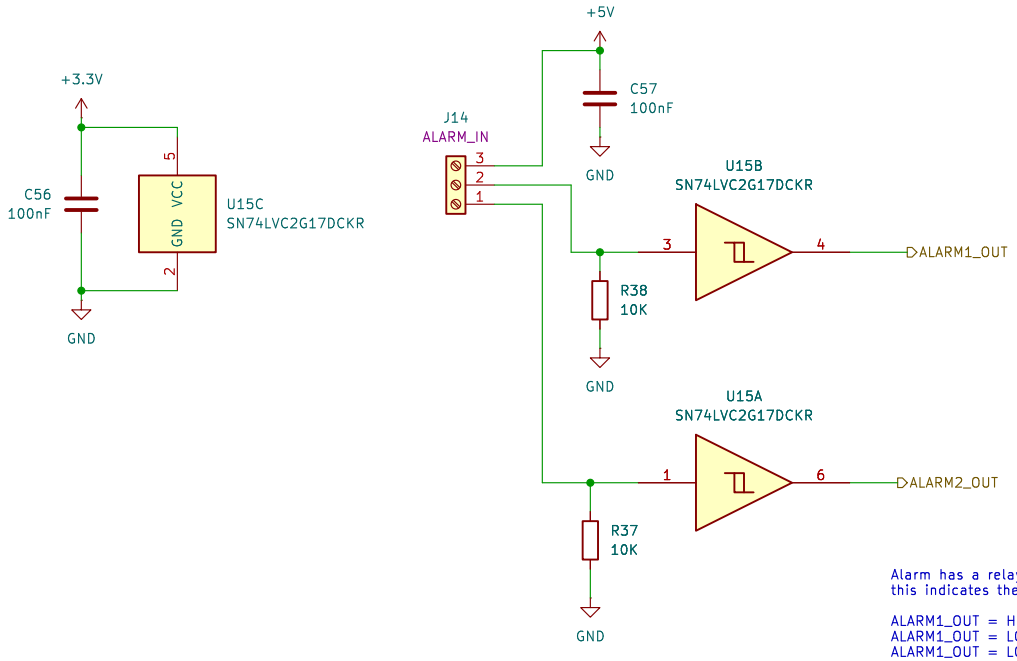



BIN/2COMP tied to DVCC (+3.3V) by default.
Input coding is set to offset binary.

| | | |
|---|---|---|
| Sheet: /DAC/<br>File: dac.kicad_sch | | |
| **Title:** | | |
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 9.0.3 | | Id: 9/14 |

# Chiller alarm inputs

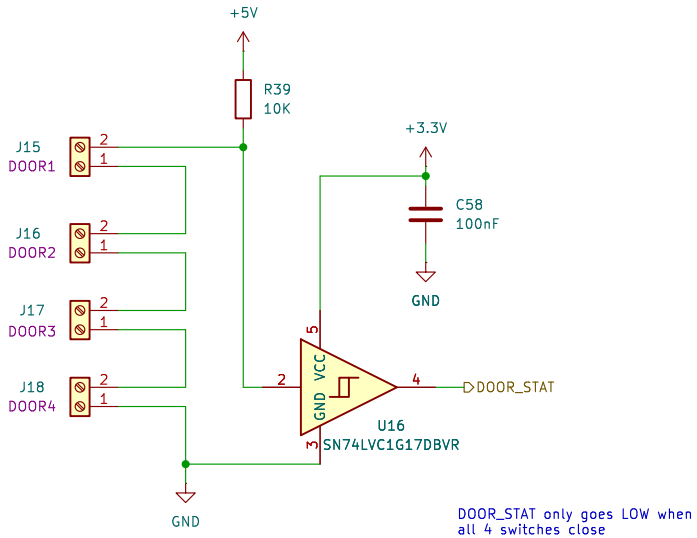



Alarm has a relay to short its terminals to 5V,
this indicates the alarm state.

ALARM1_OUT = HIGH, ALARM2_OUT = LOW : Alarm active
ALARM1_OUT = LOW, ALARM2_OUT = HIGH : Alarm disabled
ALARM1_OUT = LOW, ALARM2_OUT = LOW : Undefined

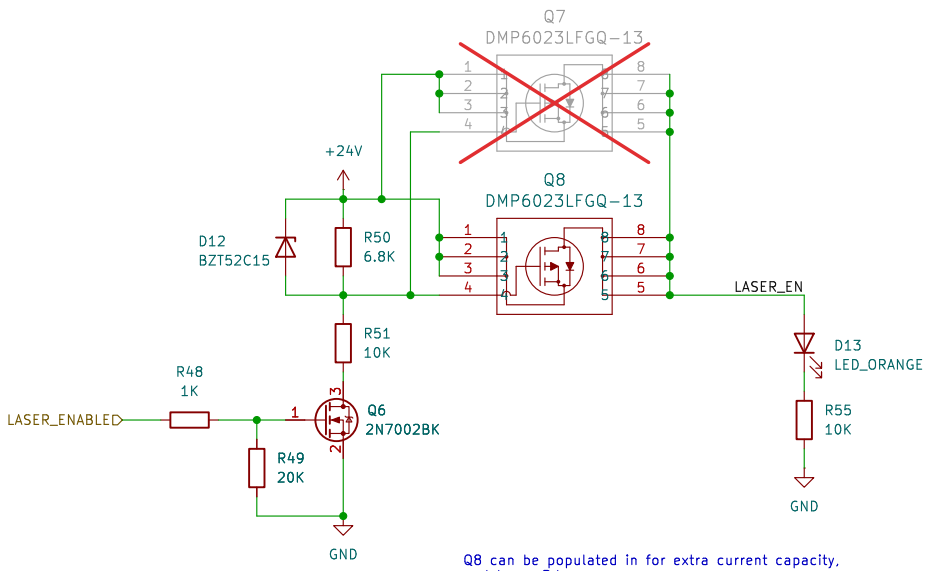
| | | |
|---|---|---|
| Sheet: /CHILLER_ALARM/ | | |
| File: chiller_alarm.kicad_sch | | |
| **Title:** | | |
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 9.0.3 | | Id: 7/14 |

## Door limit switches



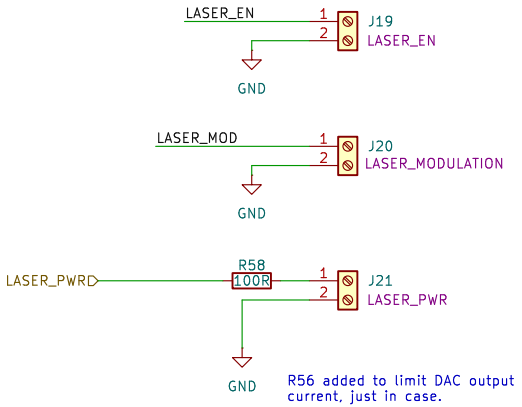

| | | |
|---|---|---|
| Sheet: /DOOR_INPUTS/ | | |
| File: door_inputs.kicad_sch | | |
| **Title:** | | |
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 9.0.3 | | Id: 8/14 |

## Laser power enable switch



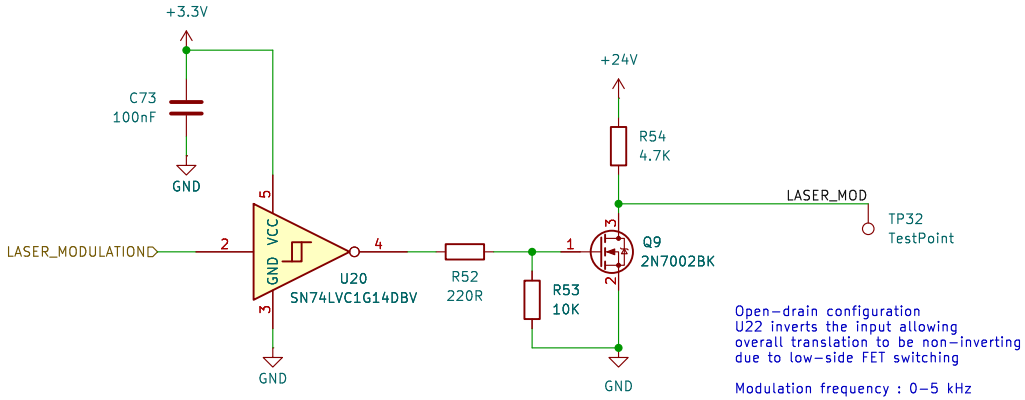

Q8 can be populated in for extra current capacity, and lower Rds on.

## Laser connectors




R56 added to limit DAC output current, just in case.

## Laser modulation signal translation 3.3->24V



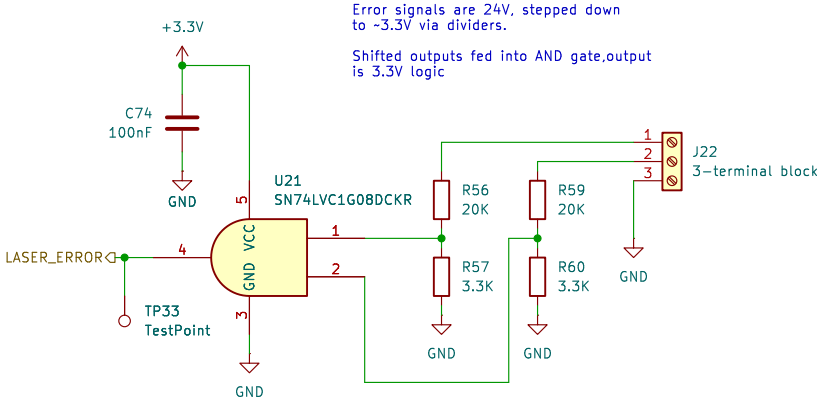

Open-drain configuration
U22 inverts the input allowing overall translation to be non-inverting due to low-side FET switching

Modulation frequency : 0-5 kHz

## Error signals AND gate

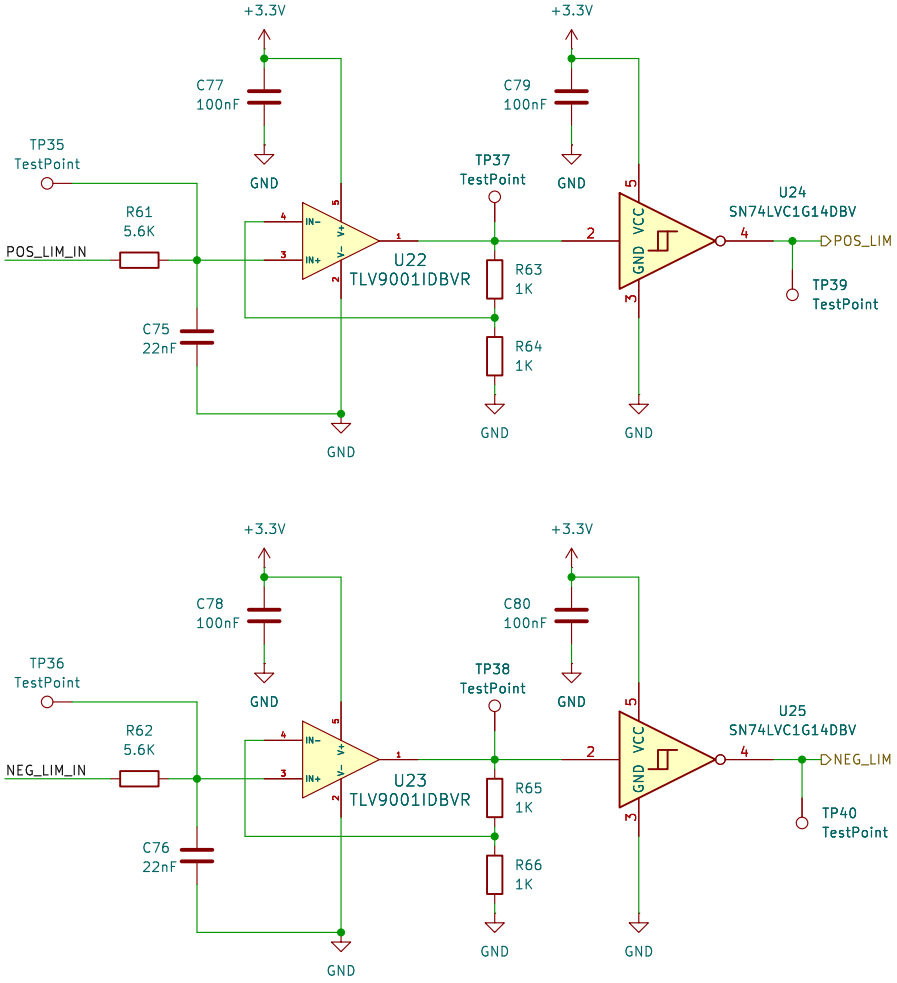
Error signals are 24V, stepped down to ~3.3V via dividers.

Shifted outputs fed into AND gate,output is 3.3V logic




| | | |
|---|---|---|
| Sheet: /LASER/<br>File: laser.kicad_sch | | |
| **Title:** | | |
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 9.0.3 | | Id: 10/14 |

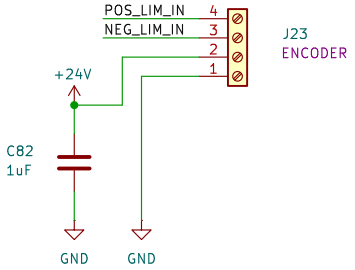
## Encoder POS & NEG lim signals active low pass filters



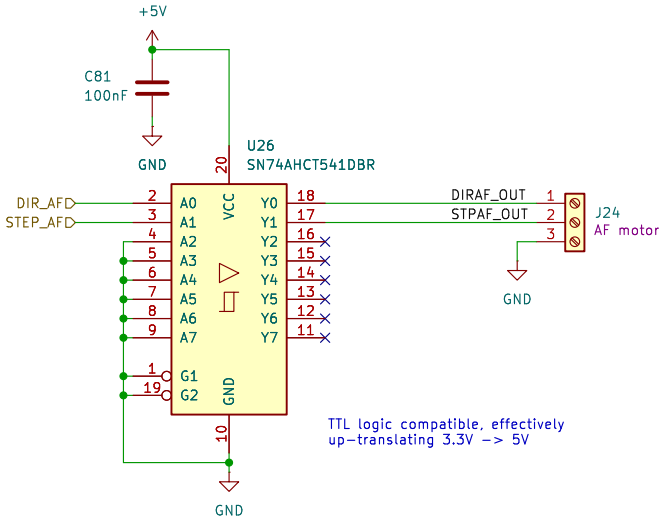

Active low pass filter cutoff
fc ~= 1290 Hz

NOT gates reverses inverted signal
from op-amp so that inputs and outputs
are not inverted.

## AF stepper encoder signals connector

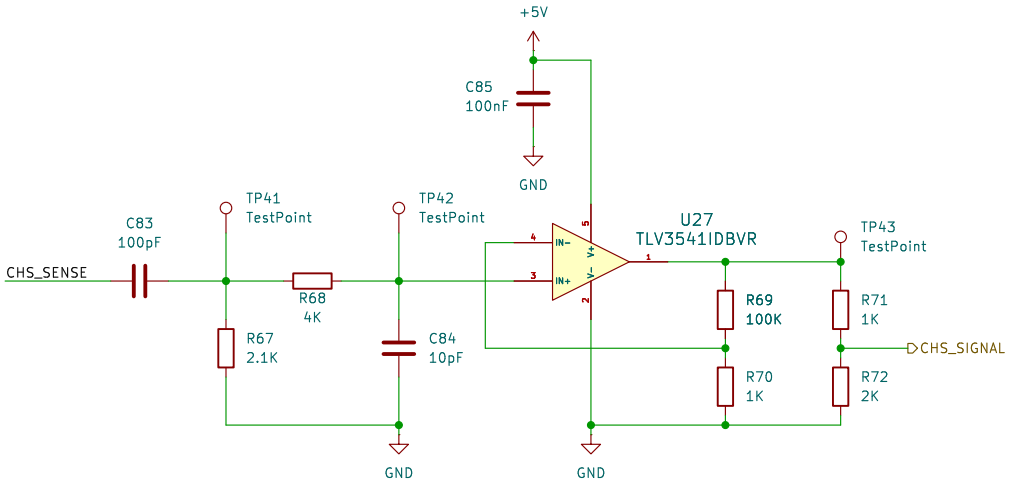



## AF stepper control signals driver




| | | |
|---|---|---|
| Sheet: /AF_STEPPER/ | | |
| File: af_stepper.kicad_sch | | |
| **Title:** | | |
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 9.0.3 | | Id: 11/14 |

## Bandpass filter for CHS sensor signal

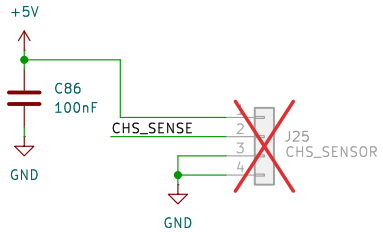



Bandpass filter with cutoff frequencies;
750kHz ~ 4MHz

Passive bandpass filter followed by op-amp
gain stage. Gain = 100

Voltage divider added to scale
down output to 3.3V range.

## CHS Sensor input




J25 should be a GX16-4 connector

| | | |
|---|---|---|
| Sheet: /CHS_SENSOR/<br>File: chs_sensor.kicad_sch | | |
| **Title:** | | |
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 9.0.3 | | Id: 12/14 |